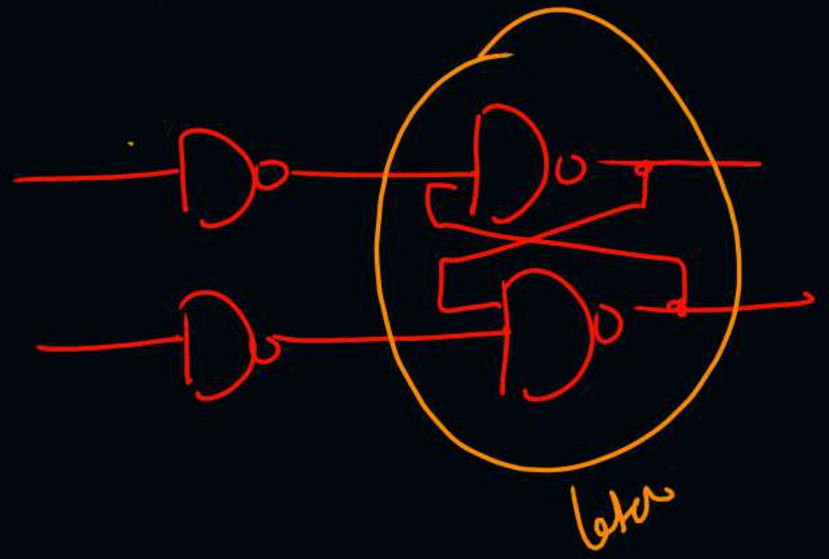
Latches → Flip Flops → Registers

(Basic building block of Flip Flops)



"Stay connected on Youtube"

Computer

35 - 40

750

SAMARTH

CUET PG 2027

Lecture – 15 [redacted]

Unit 3 : Digital Fundamental
Topic : Registers

By- Puneet Kaur



Recap of previous lecture

1. Flip Flops :- Capable of storing 1 bit of information



Topics *to be covered*

1. Registers

Morris Mano
Book

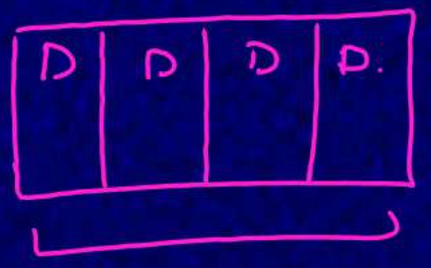


Topic : Registers

Topic : Registers

Combination of Flip Flops.

Ex 4 bit Register = 4 D flip flops



4 bit Register

What is a Register in a Computer?
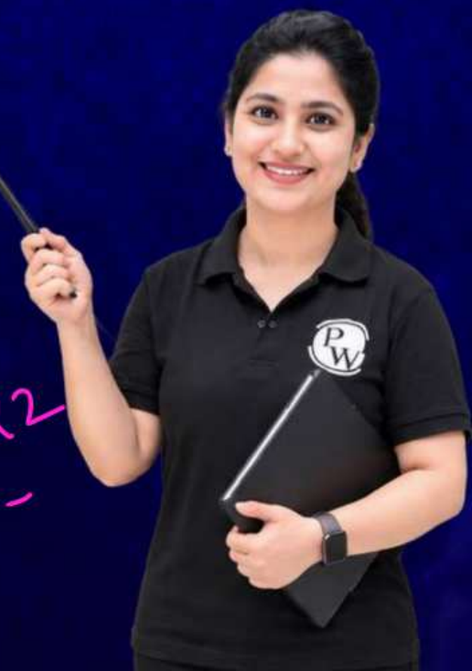
- A **register** is a small, high-speed storage unit located directly in the **CPU**. Registers temporarily store data, instructions, and addresses that the CPU needs during the execution of instructions. They are the **fastest memory elements** in the memory hierarchy and serve as the CPU's **working memory**.

CPU Registers

- A register can hold a **bit sequence**, an instruction, a memory address, or any other data that supports fast, real-time computation.

Register = R

R0 R1 R2 ...



Topic : Registers
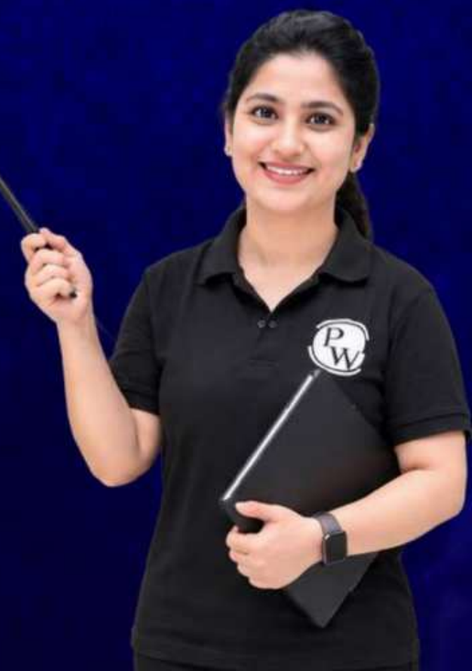
Why Registers Are Important

- **Faster than cache and RAM**
- Help **reduce memory access time**
- Store intermediate and final computation results
- Used for **instruction execution, data manipulation**, and **status tracking**

To be discussed later also.

CPU
Registers
Cache
RAM
SM

Speed ↓, Access time ↑

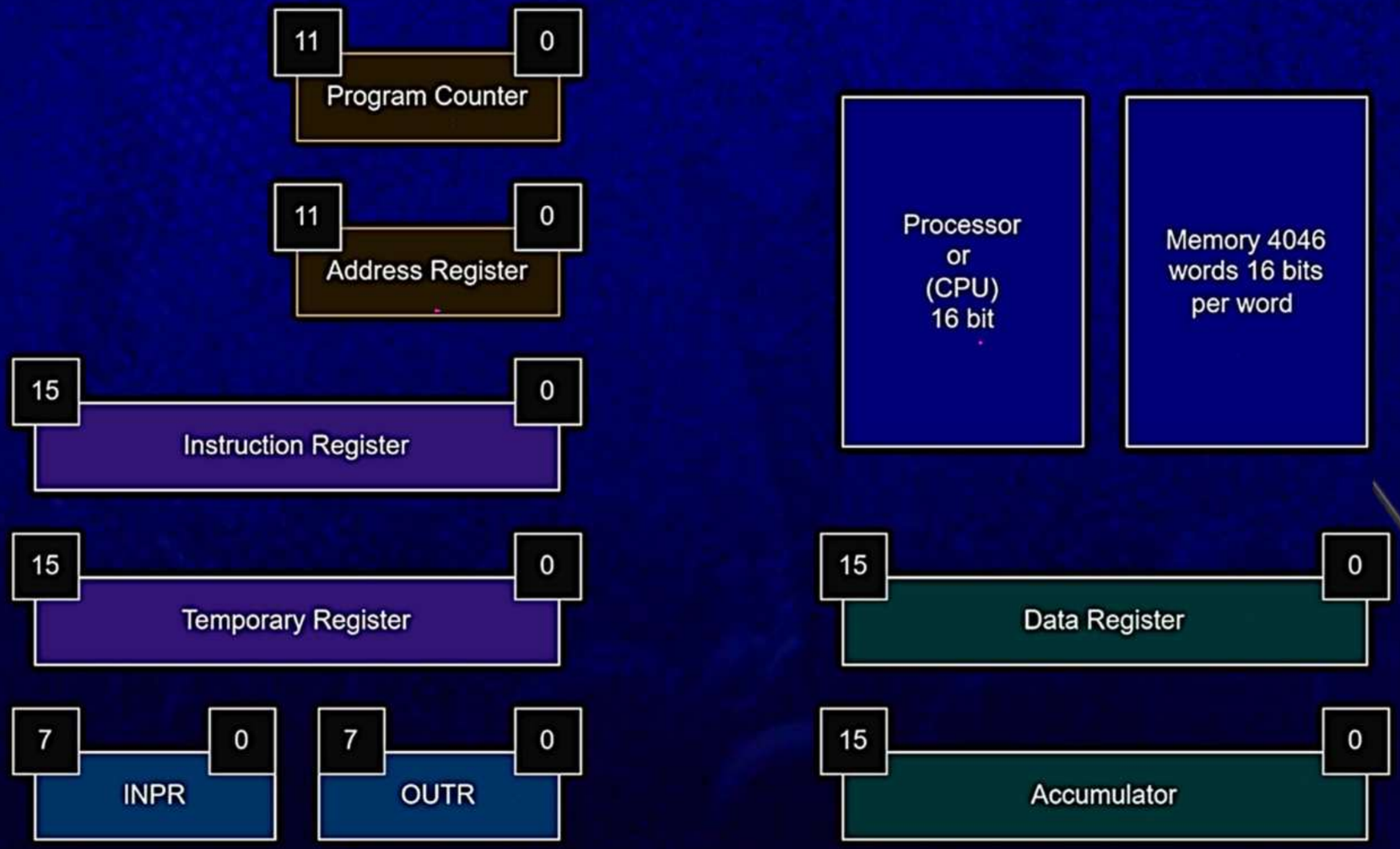


Topic : Registers

Basic computer ✓

Expected imp

Categories / Classes of CPU Registers

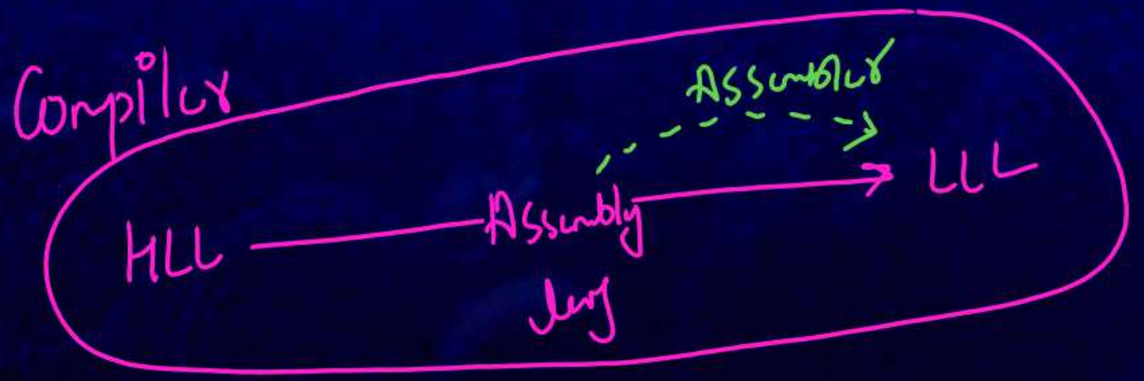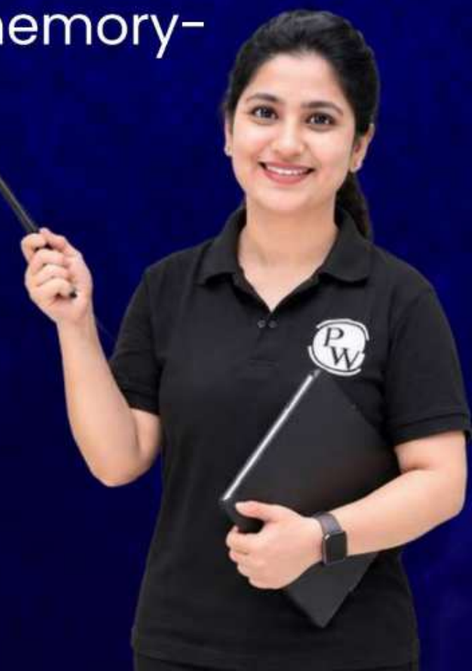




Topic : Registers

1. **General Purpose Registers (GPRs)**

- **Symbol**: R0, R1, R2, ..., Rn
- Used to store **temporary data, intermediate results**, and values during program execution.
- **Accessed by** assembly-level instructions.
- Widely used in **register-to-register operations** (faster than memory-based operations).





Topic : Registers

Also called as Processor Register

2. Accumulator Register (AC)
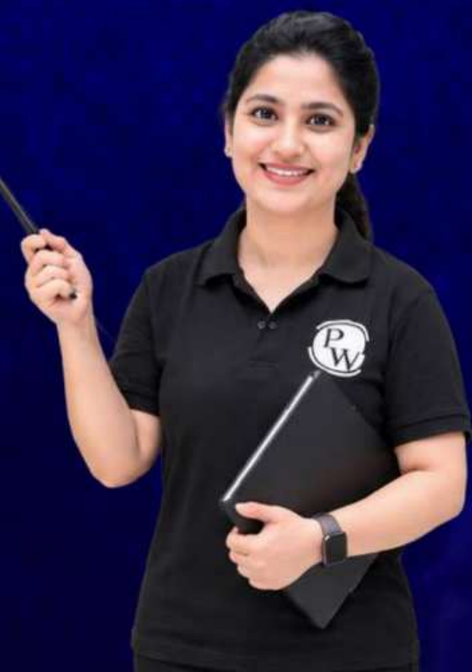
- Stores **intermediate arithmetic and logic results** from the ALU.
- Acts as a **default operand/result location** if not explicitly mentioned in the instruction.
- Frequently used in older or simpler CPUs.

AC ← 16 x 2

AC ← AC + 8

8 + (16 x 2)

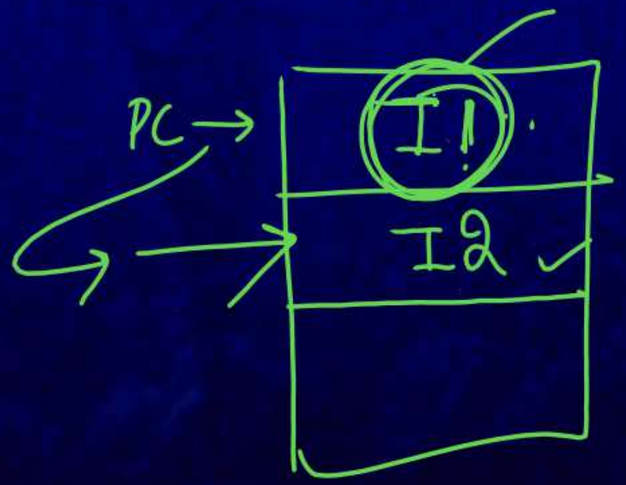
AC
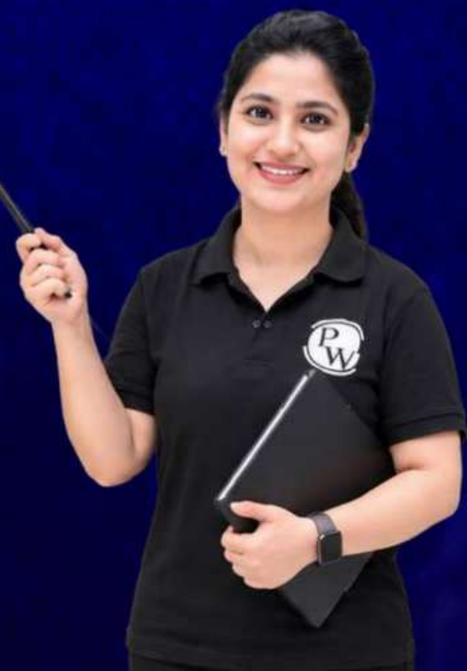


Topic : Registers

3. Program Counter (PC)

- Holds the **address of the next instruction** to be executed.
- **Automatically incremented** after each fetch cycle.
- Size depends on architecture (e.g., 12-bit PC can address 4K memory locations).
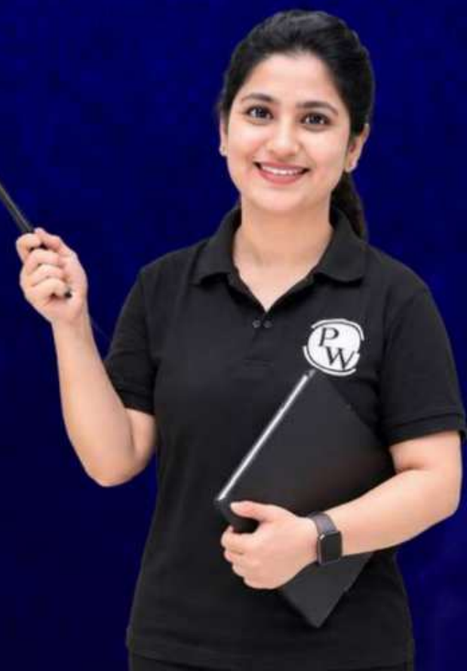




Topic : Registers

4. Instruction Register (IR)

- Holds the **current instruction** fetched from memory.
- The instruction is **decoded and executed** from here.
- Number of bits = number of bits in an instruction.

IR — Current class — Puneet Maam

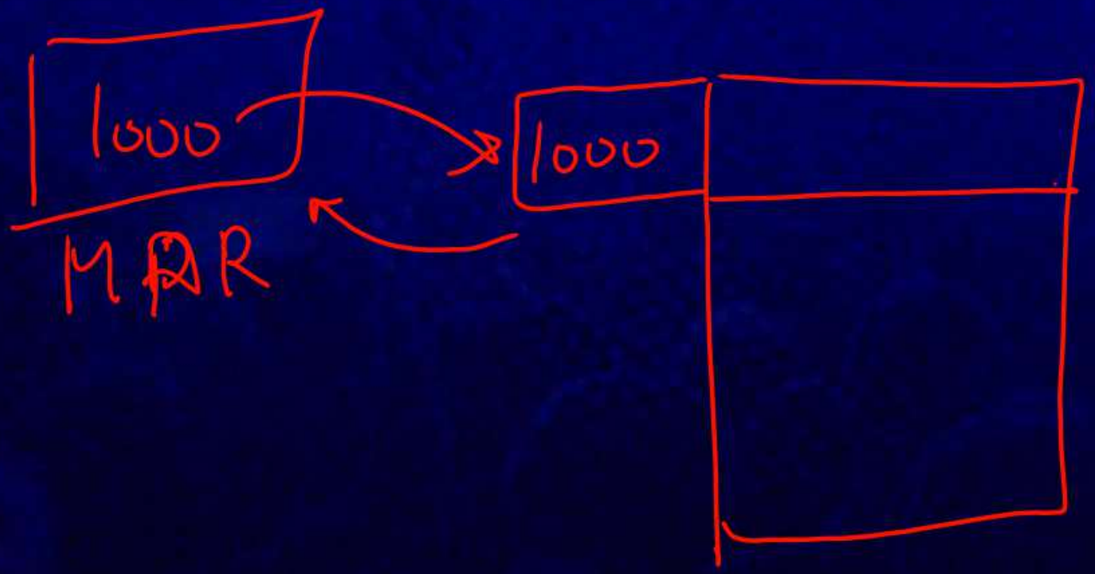
PC — Next class — Kotlan sir
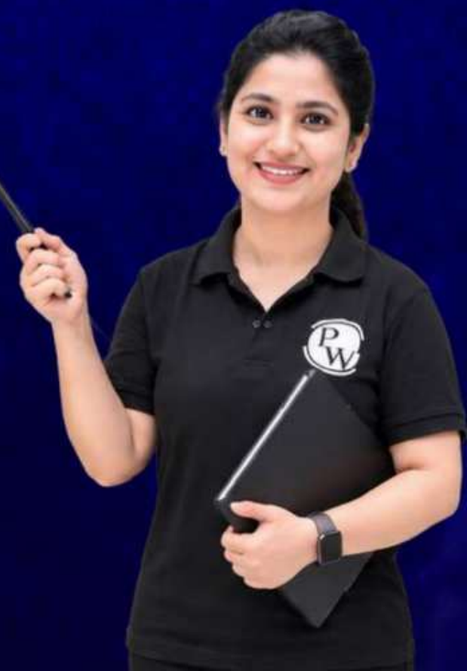


Topic : Registers

5. Memory Address Register (MAR)

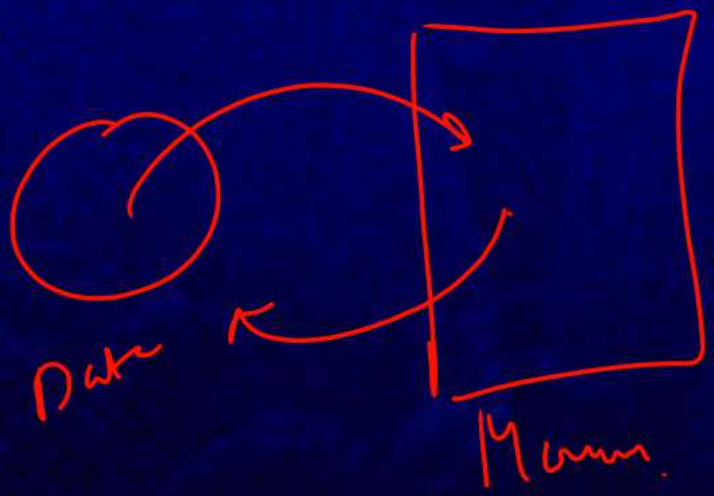
- Holds the **memory address** of the data or instruction to be fetched or stored.
- Used alongside **MDR** for memory operations.
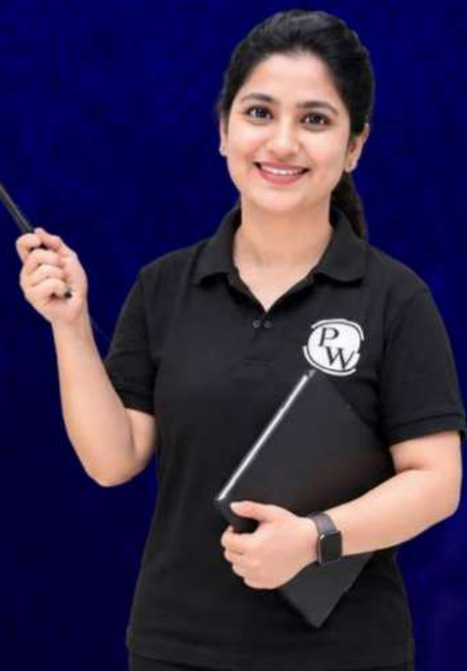




Topic : Registers

6. **Memory Data Register (MDR) / Memory Buffer Register (MBR)**
   - Temporarily stores **data being transferred to/from memory**.
   - Acts as a **data interface** between memory and CPU.

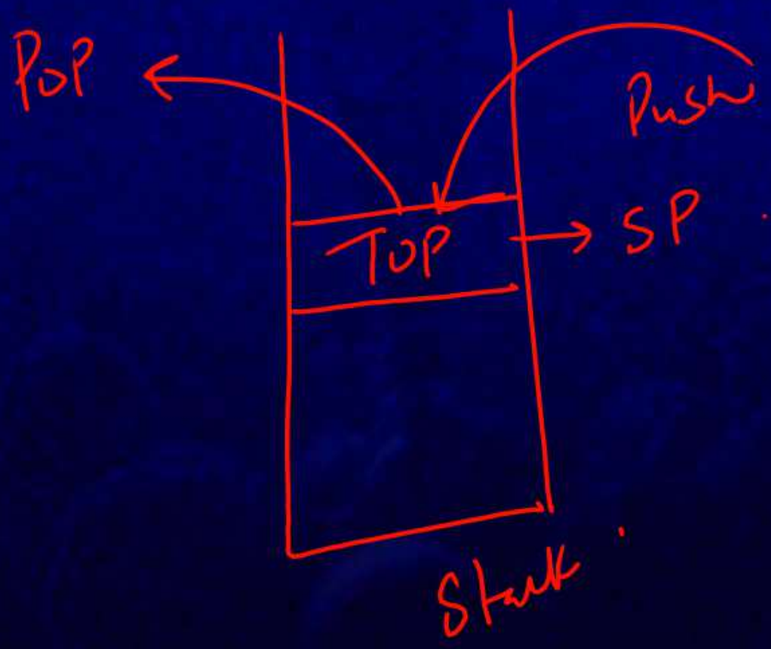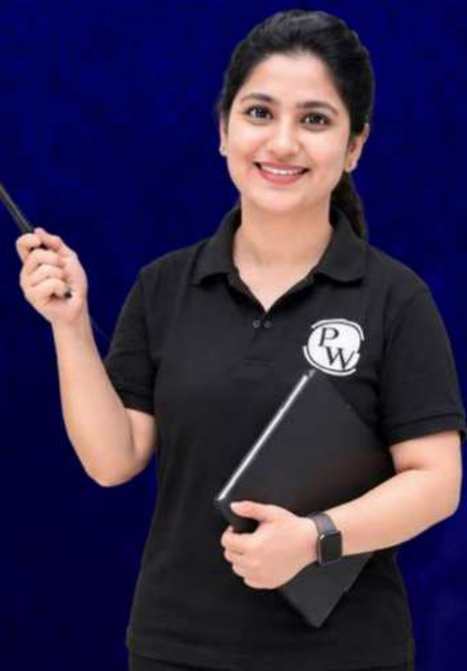




Topic : Registers

7. Stack Pointer (SP)

- Points to the **top of the stack** in memory.
- Stack is used for **function calls, interrupts, and recursive procedures**.
- SP updates automatically on PUSH and POP.
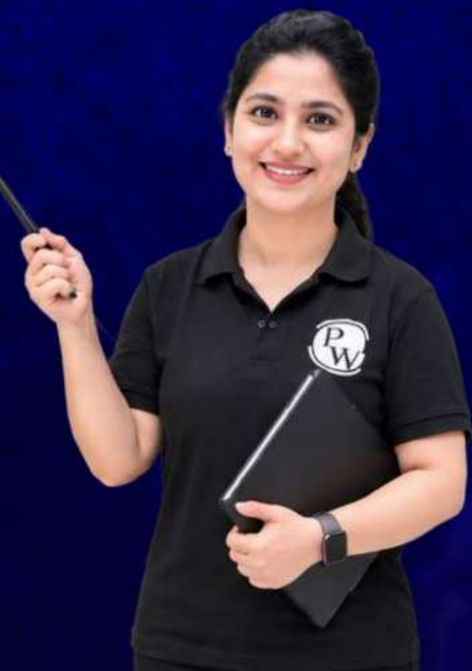




Topic : Registers

8. Flag Register / Status Register / Condition Code Register (CCR)
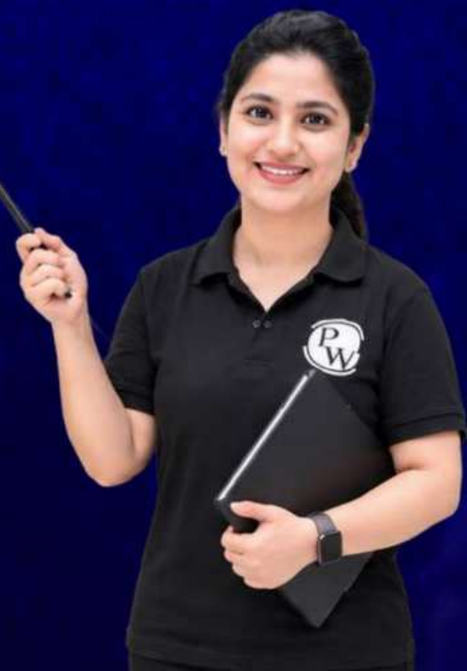
- Stores **binary flags** that describe the **outcome of operations**.
- Common flags:
  - **Carry (C):** Set if there is a carry/borrow.
  - **Zero (Z):** Set if the result is 0.
  - **Sign (S) or Negative (N):** Set if result is negative.
  - **Overflow (V):** Set on signed overflow.
  - **Parity, Auxiliary Carry, Interrupt Enable**, etc.
- Flags are used in **branching and conditional logic**.



Topic : Registers

9. Temporary Register (TR)

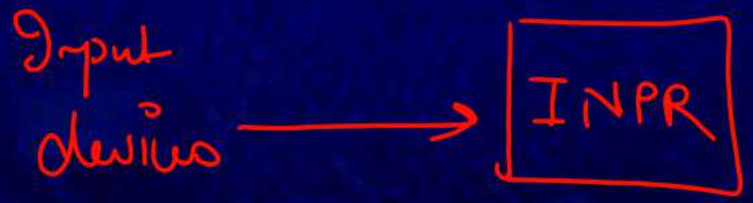
- Holds **temporary intermediate data** during complex CPU operations.
- Not programmer-accessible, used internally by control unit.

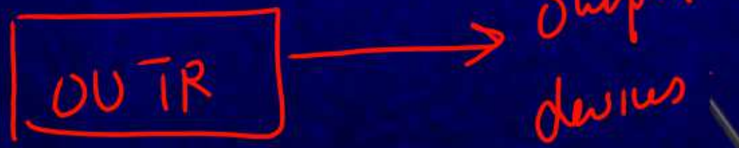


Topic : Registers

10. Input Register (INPR) and Output Register (OUTR)

- **INPR** stores characters from input devices.
- **OUTR** holds characters to be sent to output devices.
- Typically **8-bit registers**, depending on character encoding.







Topic : Registers
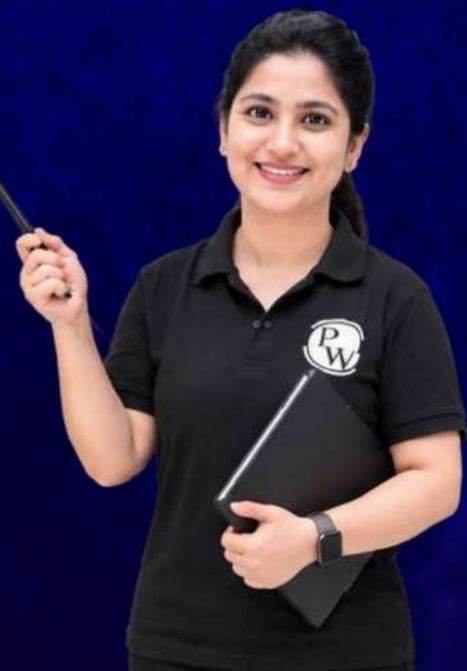
Common Registers and Their Properties

Register Name	Symbol	Bits	Function Description
Accumulator	AC	16	Arithmetic/logic operation result storage
Program Counter	PC	12	Address of next instruction
Address Register	AR	12	Holds address of data/instruction
Data Register	DR	16	Holds operand/data fetched from memory
Instruction Register	IR	16	Stores current instruction being executed
Temporary Register	TR	16	Holds temporary values during processing
Input Register	INPR	8	Stores input character from device
Output Register	OUTR	8	Stores output character to device

Topic : Registers

How Registers Work with Other CPU Components

- Registers communicate with:
  - **ALU** for arithmetic and logical operations
  - **Control Unit** for instruction execution control
  - **Memory** for data/instruction fetch and storage
  - **I/O devices** for data transfer (INPR/OUTR)
- Registers enable **fast access and processing**, reducing reliance on slower RAM.
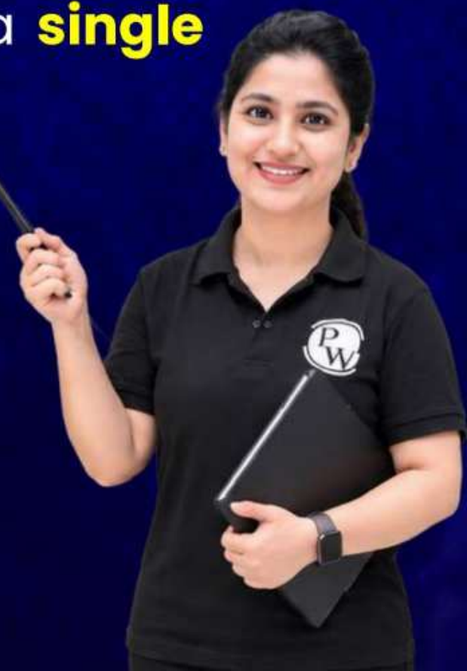


Topic : Registers

For your knowledge

Common Bus System

- A **common bus system** is a technique used to **connect multiple registers, memory, and the ALU** using a **single shared communication pathway**, called the **bus**.
- A basic computer typically consists of **8 registers**, a **memory unit**, an **adder and logic circuit**, and a **control unit**, all connected via a **single common bus**.



Topic : Registers

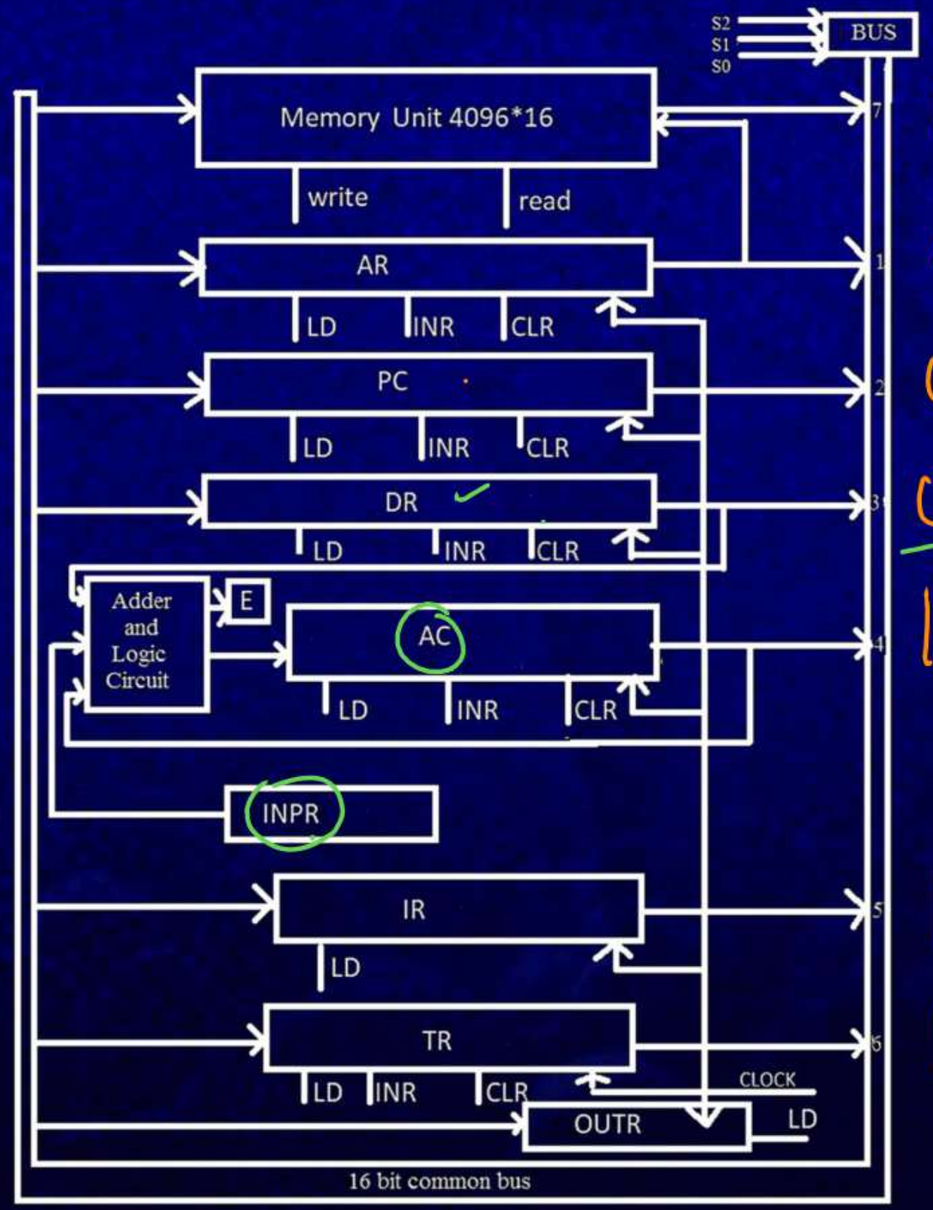
Connections in the Bus System

- All **register outputs** (except **OUTR**) are connected to the **common bus**.
- **Selection of output** to be placed on the bus is done using **control signals**: S2, S1, and S0 (3-bit binary selector).
- The **common bus lines** are connected to the **inputs of all registers and memory**.
- A register **loads data from the bus** when its LD (Load) signal is **activated**.
- The **memory** accepts data from the bus when its **Write** signal is enabled.
- The **memory places data on the bus** when its **Read** signal is activated.

Topic : Registers

Select lines
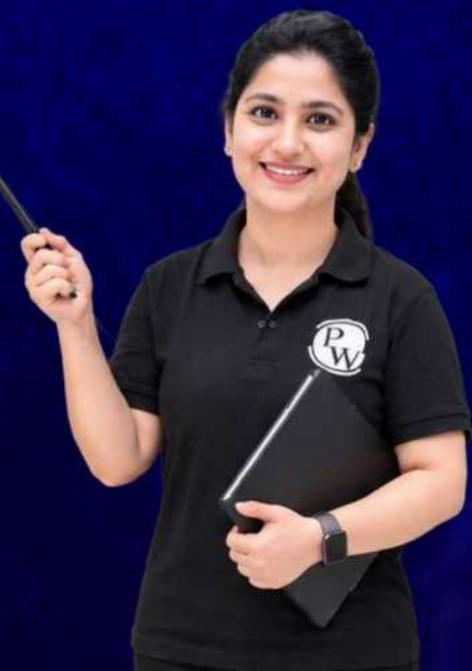
S2	S1	S0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1



Ex Memory → AC

① Memory → DR

② DR → AC
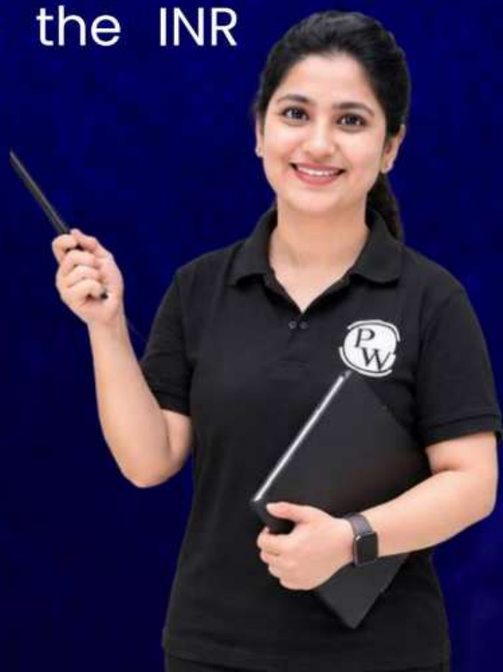


Topic : Registers

Control Inputs of Registers

Most registers (AR, PC, DR, AC, TR) have **three control inputs**:
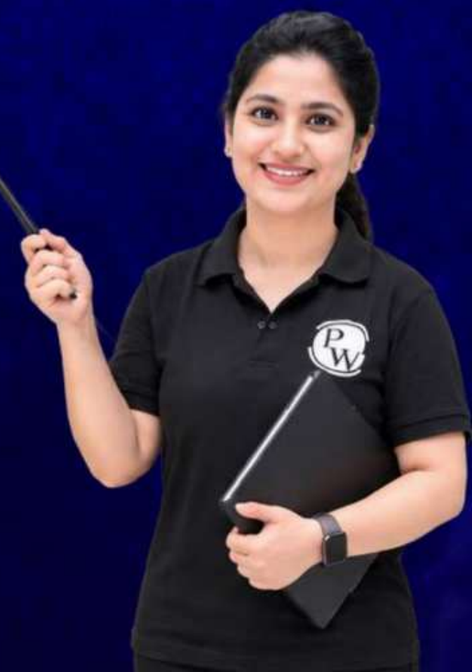
- **LD (Load)**: Load data from the bus
- **INR (Increment)**: Increment register content
- **CLR (Clear)**: Clear register content to 0
  - These registers behave similarly to binary counters due to the INR control line.



Topic : Registers

Adder and Logic Circuit

- The **Adder and Logic Unit (ALU)** provides **16-bit input** to the **AC (Accumulator)**.
- It has **three input sources**:
  1. **From AC**: For performing register operations
  2. **From DR**: For executing arithmetic and logic operations
  3. **From INPR**: For loading data from input devices
- The **result** of any operation is stored back into the **AC**.
- **E (Extend Flip-Flop)**: Holds **end-around carry** during arithmetic operations.
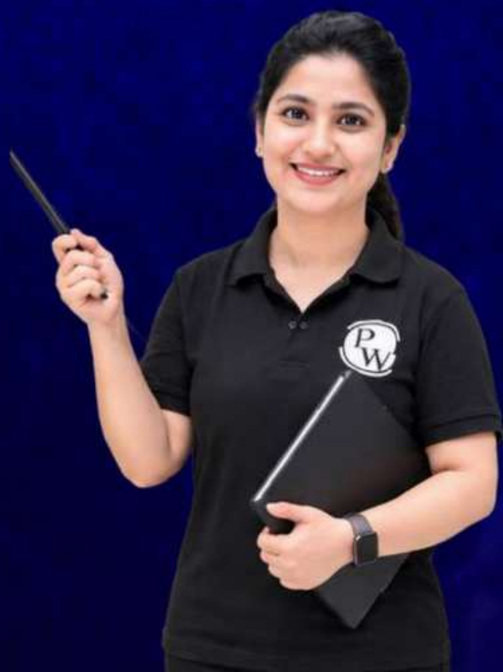


Topic : Registers

Note:

During a **single clock cycle**, the contents of any register can be:
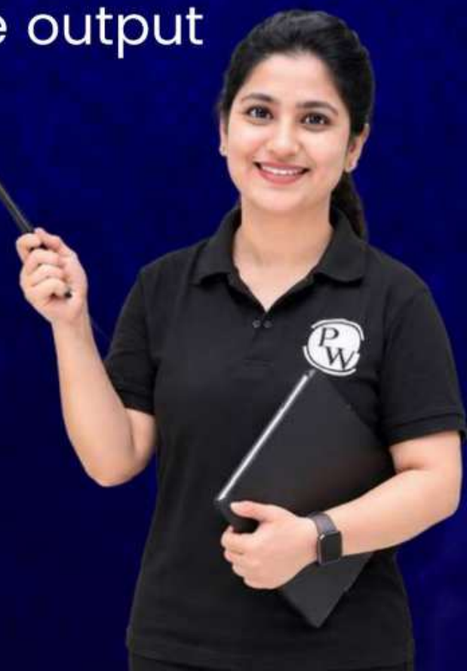
- Placed onto the **common bus**, and
- **Processed by the adder and logic circuit**
- Then the **result is stored back** into AC.



Topic : Registers

INPR and OUTR Behavior:

- **INPR (Input Register)**: Gets 8-bit data from an input device and passes it to the **AC**.
- **OUTR (Output Register)**: Takes 8-bit data from the **AC** and sends it to the output device.
- These registers are connected directly to I/O lines and not to the output side of the bus.



2 Mins Summary

Counter Theory

Notes to be provided on App

Regions



m n

Add lines & Data lines

ROM memory size $= 2^m \times n$

2025 Memory capacity $= 32K \times \frac{16}{n}$

$2^5 \quad 2^{10}$

$= 2^{15}$

$= 15$

2026
2025
2024

Homework from SAMARTH CUET PG 2027 Module

Remarks + DPP



Next week yt o.

Error detection & Correction

yt

THANK YOU